

Skills Summary

Polynomial Long Division

Use this reference while completing your worksheet or when studying.

What division claims. For any dividend P and divisor D , $P = D \cdot Q + R$, where the remainder R has lower degree than D . Every answer can be checked by multiplying back.

The four moves, repeated: divide leading term by leading term, multiply the whole divisor, subtract (change every sign), bring down the next term.

Stop when what is left has lower degree than the divisor, and write the answer as $Q + \frac{R}{D}$ — or just Q when R is zero.

1. Long division with a linear divisor

With a divisor of the form $x + a$, every round produces one quotient term and the algorithm ends when a constant is left. Subtracting is where signs get lost: bracket the product first, then flip every sign inside it.

Example: Divide $x^2 + 4x + 1$ by $x + 1$.

Step 1. The divisor has degree 1, so keep dividing until what is left is a constant — that constant is the remainder.

Step 2. $x^2 \div x = x$, and $x(x + 1) = x^2 + x$; subtracting leaves $3x$, then bring down 1. Next $3x \div x = 3$, and $3(x + 1) = 3x + 3$; subtracting leaves $1 - 3$.

The answer is $x + 3 - \frac{2}{x+1}$.

Try it: Divide $x^2 + 7x + 4$ by $x + 2$.

check: $\frac{x^2+x}{9} - 2 + x$

2. When a power is missing from the dividend

Long division lines terms up by degree, so a missing power must be written in with a zero coefficient before you start. Skipping it shifts every later column one place and the quotient comes out a degree short.

Example: Divide $x^3 + 3x - 4$ by $x - 1$.

Step 1. There is no square term, so rewrite the dividend as $x^3 + 0x^2 + 3x - 4$ before any dividing happens.

Step 2. $x^3 \div x = x^2$ leaves $x^2 + 3x - 4$; then $x^2 \div x = x$ leaves $4x - 4$; then $4x \div x = 4$ leaves nothing.

The remainder is zero, so the answer is $x^2 + x + 4$.

Try it: Divide $x^3 - 2x + 1$ by $x + 2$.

check: $\frac{x^2+x}{6} - 2 + x$

3. Dividing by a quadratic divisor

Nothing about the four moves changes — only the stopping rule does. With a degree-2 divisor the remainder may still carry an x , so the division ends as soon as the leftover has degree 1 or 0.

Example: Divide $x^3 + 2x^2 + 3x + 4$ by $x^2 + 1$.

Step 1. The divisor has degree 2, so stop as soon as what is left has degree below 2 — a linear leftover is a legitimate remainder here.

Step 2. $x^3 \div x^2 = x$ leaves $2x^2 + 2x + 4$; then $2x^2 \div x^2 = 2$ leaves $2x + 2$, which has degree 1 and stops the algorithm.

The answer is $x + 2 + \frac{2x+2}{x^2+1}$.

Try it: Divide $x^3 + x^2 + 2x + 5$ by $x^2 + 2$.

check: $\frac{x^3+x^2}{x^2+2} + 1 + x$

(!) **Watch out:** two errors account for most lost marks. The first is subtracting only the first term of the product instead of all of it. The second is stopping too early or too late — compare the degree of what is left with the degree of the divisor after every round, and let that decide.